

K-FIELD LUCAS SUPERGROUP ANALYSIS

An exhaustive foundational reference on the 9B block containing 729 Cell3 units: complete cellular topology, polarity grading, geometric derivatives, ternary phase extension, laminar and graph spectra, reciprocal primitive directions, and hidden metric dimensions

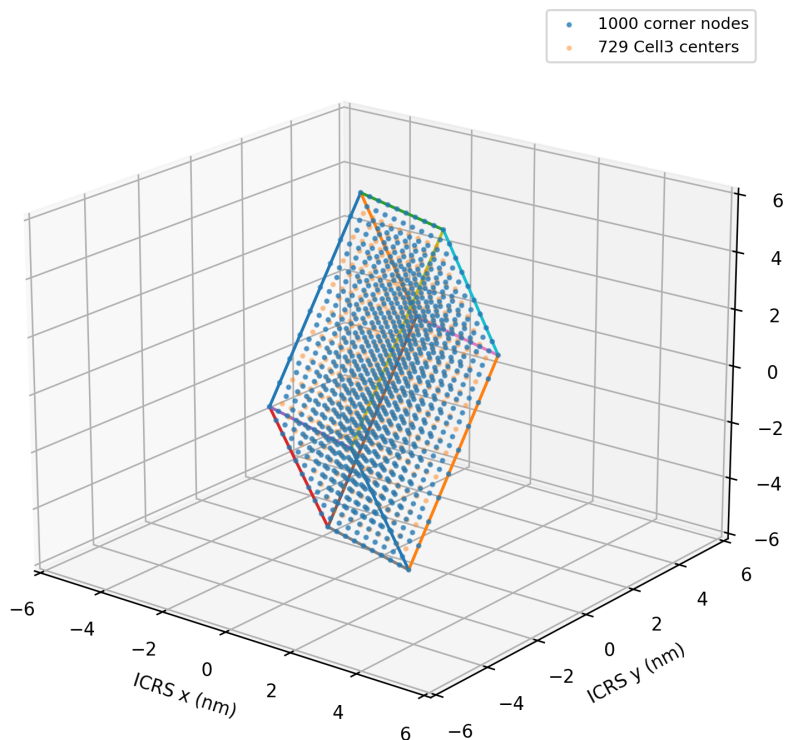
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Krytronx

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Mass-Bridge completion block: $9 \times 9 \times 9$ Cell3 volumes



Title figure. Dimensionally correct ICRS rendering of the Lucas Supergroup. The $9 \times 9 \times 9$ space-filling block contains 729 complete Cell3 units, 27 Lucas Groups, and 1,000 merged unrefined corner nodes. All plotted coordinates are generated from the locked Cell3 basis.

Archival computational edition - Unix epoch 1785208014

Document Control

Field	Value
Title	K-Field Lucas Supergroup Analysis
Release type	Foundational educational and archival computational reference
Principal Investigator	P. Dwayne Esterline
Organization	Krytronx
Epoch identifier	1785208014
UTC generation time	2026-07-28T03:06:54+00:00
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Primary locked source	K-Field_Mass-Bridge_is_now_a_locked_projective_ratio_system_1785176592(1).json
Prior hierarchy source	K-Field_mass_bridge_drives_volume_filling_hierarchy_1785204375.json
Coordinate frame	ICRS Cartesian; locked Cell3 basis vectors form the columns of B
Units	nm, nm ² , nm ³ , and reciprocal nm ⁻² unless stated otherwise
Machine-readable companion	K-Field_Lucas_Supergroup_Analysis_1785208014.json

The term Lucas Supergroup denotes the 9B space-filling block containing $9^3=729$ complete Cell3 units, equivalently a 3 x 3 x 3 space-filling arrangement of 27 Lucas Groups. This definition is used consistently throughout the report.

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Abstract

The Lucas Supergroup is the 9B volume-filling block formed from 729 locked Cell3 units. The Mass Bridge independently selects this level because a row of nine cells has ten corner-node positions, a primitive face has one hundred corner-node positions, and the full corner lattice has one thousand positions. Those counts multiply the locked face ratios into η alpha, α^{-1} , and η . The Supergroup is therefore the minimal recursive volume in which the complete Mass-Bridge dimensional composition is represented by the lattice itself.

Exhaustive analysis reveals that the $19^3=6,859$ half-step nodes are in one-to-one correspondence with every vertex, edge, face, and Cell3 volume in the complete $9 \times 9 \times 9$ cubical complex. The resulting cellular generating polynomial is $F(t)=(10+9t)^3$. Its value $F(1)=6,859$ gives the total barycentric state count, while $F(-1)=1$ gives the Euler characteristic of the filled solid. Checkerboard polarity is therefore the alternating dimension grading of the cellular complex. The boundary complex has Euler characteristic two, as expected for a topological two-sphere.

The Supergroup also exposes exact geometric derivative relations, consecutive cubic ladders, a two-level balanced-ternary address, a nonsplit Z_9^3 phase extension, 55 laminar sheets, a 55-dimensional adjacency nullspace, 1,216 center-graph circulation dimensions, 864 new primitive antipodal outer-shell directions, and 858 reciprocal q-compatibility dimensions. The primitive reciprocal observation system decomposes the six-dimensional symmetric metric into a one-dimensional trace scalar, a two-dimensional A2 diagonal-shape sector, and a three-dimensional shear sector. Three exact primitive integer addresses closely approximate the continuous reciprocal principal axes.

The strongest physical correspondence remains the electron-to-shielded-helion magnetic-moment ratio 864.058255 versus the exact Supergroup primitive-direction count 864, with a residual of 67.420 ppm. The geometric count is exact; the physical identification remains a forward-test candidate requiring a mechanism and broader predictive closure.

1. Reader Orientation and Analytical Discipline

This paper separates source data, exact derivation, structural interpretation, and physical correspondence. The locked Cell3 basis, metrics, face ratios, and 27-node grammar are source data. Counts, topological identities, spectra, and integer-address results are exact or directly computed derivations. Interpretations describe how those identities organize the hierarchy. Physical correspondences compare exact Supergroup quantities with measured constants and are not used to alter the geometry.

Statement class	Definition	Examples
Locked source	Numerical or structural object supplied by the authoritative files	B, M, R, face ratios, 27 nodes, six boundary planes
Exact derivation	Algebraic or enumerative consequence of the locked objects	f-vector, Euler characteristics, sheet populations, shell counts
Computed spectrum	Eigenvalue or rank calculation from an explicitly defined finite matrix	Emily nullities, q observation rank, metric-sector singular values
Structural interpretation	Meaning assigned to an exact relationship	polarity as cellular grading; object-to-field transition
Physical correspondence	Comparison with an independently measured quantity	864 primitive directions versus 864.058255

Central finding: The Lucas Supergroup is the first complete cellular, topological, spectral, and reciprocal closure of the locked Cell3 geometry.

2. Locked Cell3 Geometry

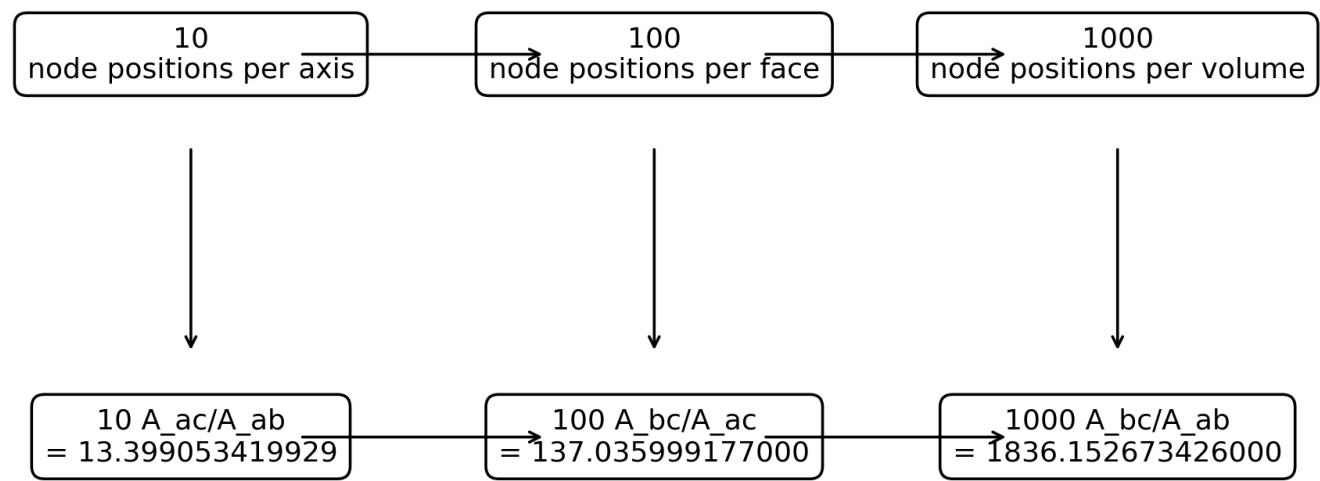
2.1 Mass-Bridge shape forcing

The three primitive Cell3 face areas obey the locked projective relations

$$\begin{aligned}
 10 \ A_{ac}/A_{ab} &= 13.399053419929 = \eta \ \alpha \\
 100 \ A_{bc}/A_{ac} &= 137.035999177000 = \alpha^{-1} \\
 1000 \ A_{bc}/A_{ab} &= 1836.152673426000 = \eta
 \end{aligned}$$

The ratios fix the projective shape point. Only a common positive scale remains free. Consequently every accumulated Supergroup ratio, angle, normalized projection, parity rule, and reciprocal anisotropy follows from one locked local geometry.

Mass-Bridge dimensional closure selected by the g=2 node skeleton



Exact closure: $(10\ r_1)(100\ r_2) = 1000(r_1\ r_2)$, so $(\eta\ \alpha)(\alpha^{-1}) = \eta$

Figure 2.1. Exact Mass-Bridge dimensional closure at the Lucas Supergroup scale. Ten line positions, one hundred face positions, and one thousand volume positions multiply the three locked face ratios into $\eta\ \alpha$, α^{-1} , and η .

2.2 Numerical kernel

Measure	Locked value
a	0.334179679059111 nm
b	0.586410890411141 nm
c	0.792319765042685 nm
theta_ab	83.933483012313 deg
theta_ac	80.448130018338 deg
theta_bc	50.363231813756 deg
A_ab	0.194869165460484 nm^2
A_ac	0.261106235790205 nm^2
A_bc	0.357809539128562 nm^2
Volume	0.117914891911745 nm^3
Body diagonal	1.343653621675925 nm

$$M = B^T B$$
$$R = M^{-1} = B^{-1} B^{-T}$$
$$||B\ u||^2 = u^T M\ u$$
$$||B^{-T}\ q||^2 = q^T R\ q$$

Locked Cell3: 27-node midpoint grammar and authoritative edge basis

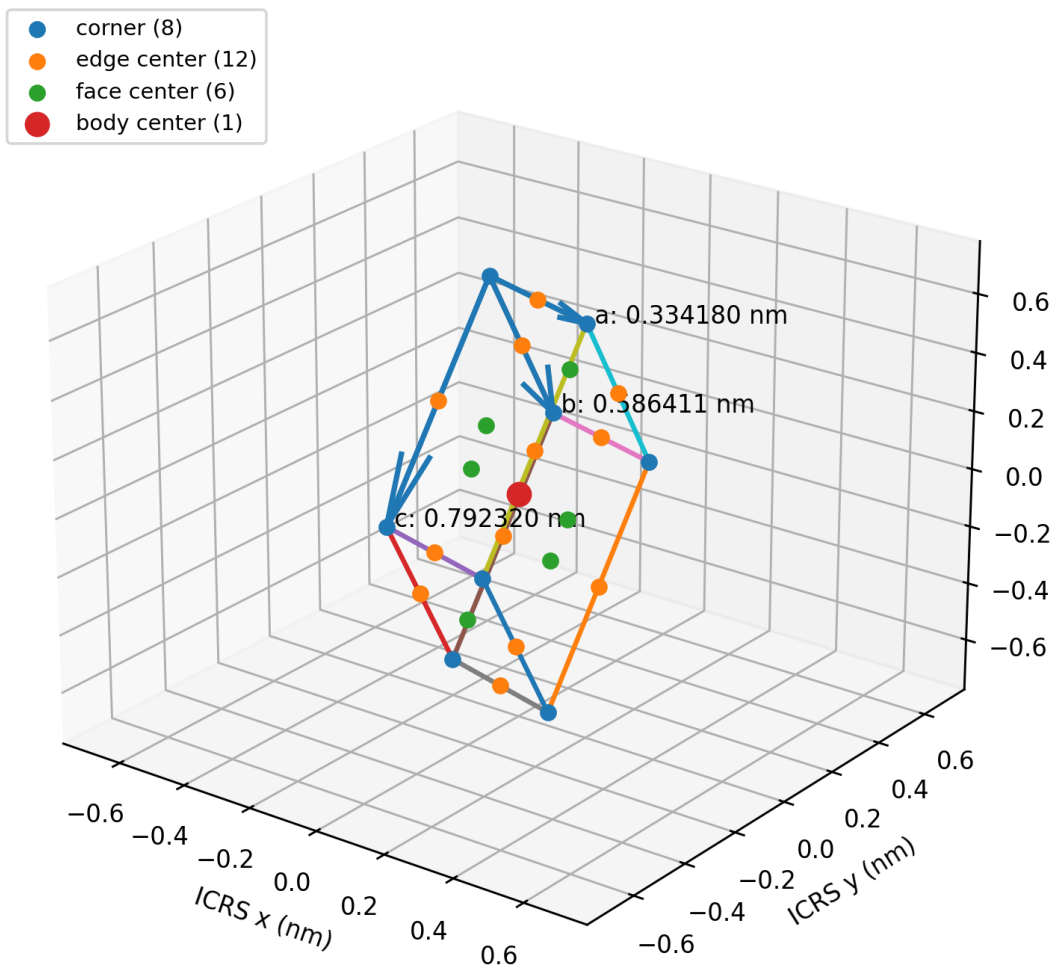


Figure 2.2. Dimensionally correct locked Cell3 with all 27 midpoint nodes, enclosing edges, and ordered primitive basis vectors.

Complete Cell3 address atlas: all 27 nodes, lambda values, and polarities

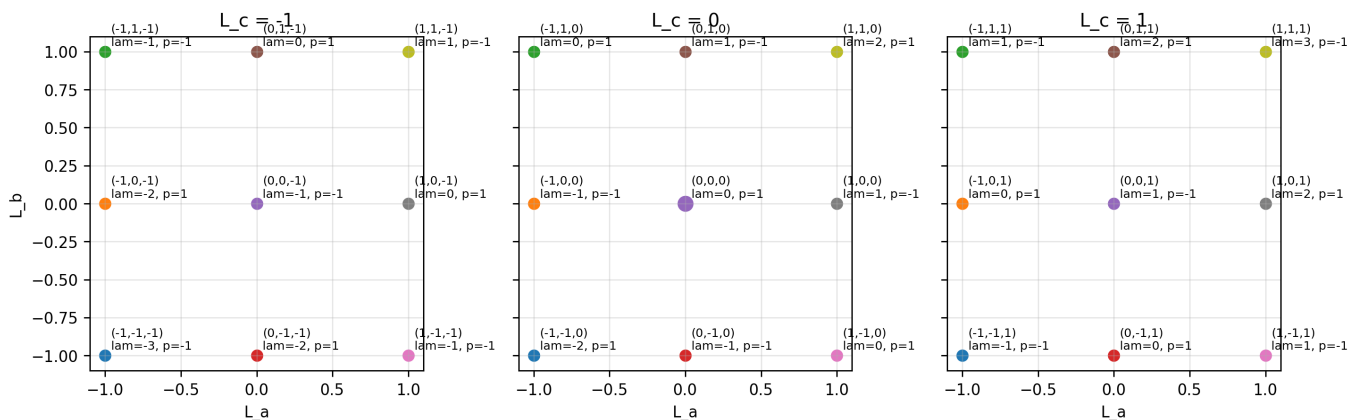


Figure 2.3. Complete 27-node address atlas. Every local node is identified by its primitive integer address, lambda value, class, and polarity.

2.3 Six boundary planes

Channel	Face family	Unit normal in ICRS	Area (nm ²)	Spacing (nm)
a	bc	-0.995965190, 0.083602163, -0.032619291	0.357809539129	0.329546529695
b	ac	-0.089052594, -0.866576769, 0.491033948	0.261106235790	0.451597379721
c	ab	0.194092201, 0.913177169, 0.358379235	0.194869165460	0.605097741519

Authoritative Cell3 boundary-plane atlas: six faces, nine nodes per face

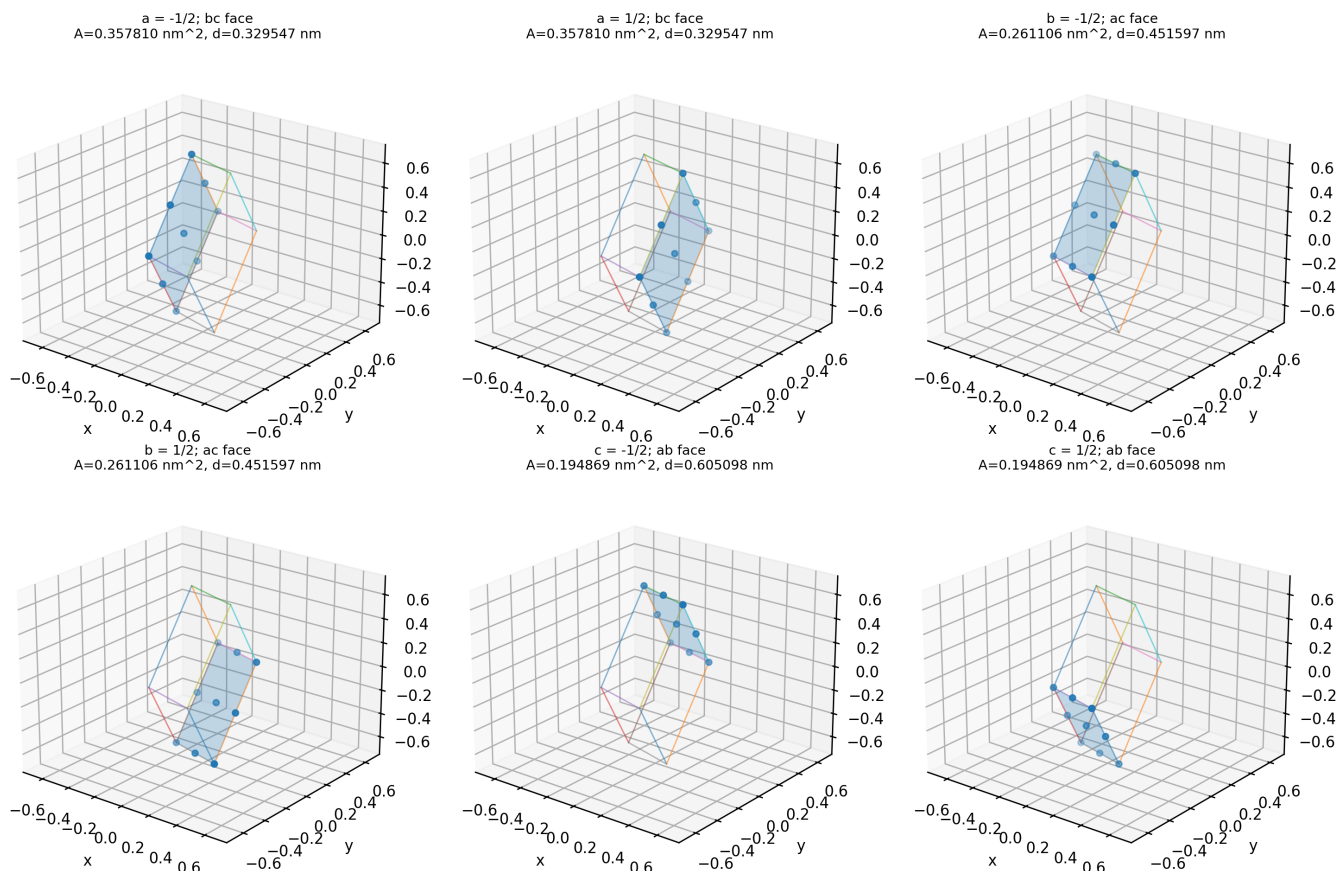


Figure 2.4. Dimensionally correct six-plane atlas. Each panel shows one boundary face, its nine local nodes, unit normal, area, and opposite-face spacing.

3. Construction and Complete Inventory of the Lucas Supergroup

3.1 Space-filling hierarchy

A Lucas Group contains three complete Cell3 units along each primitive direction and therefore has outer basis 3B and 27 Cell3 volumes. A Lucas Supergroup contains three complete Lucas Groups along each primitive direction. Its outer basis is 9B and its volume contains 27 Lucas Groups or 729 Cell3 units.

```

B_Lucas = 3B
B_Supergroup = 3(3B) = 9B
N_Cell3 = 9^3 = 729
N_LucasGroups = 3^3 = 27

```


Mass-Bridge completion block: 9 x 9 x 9 Cell3 volumes

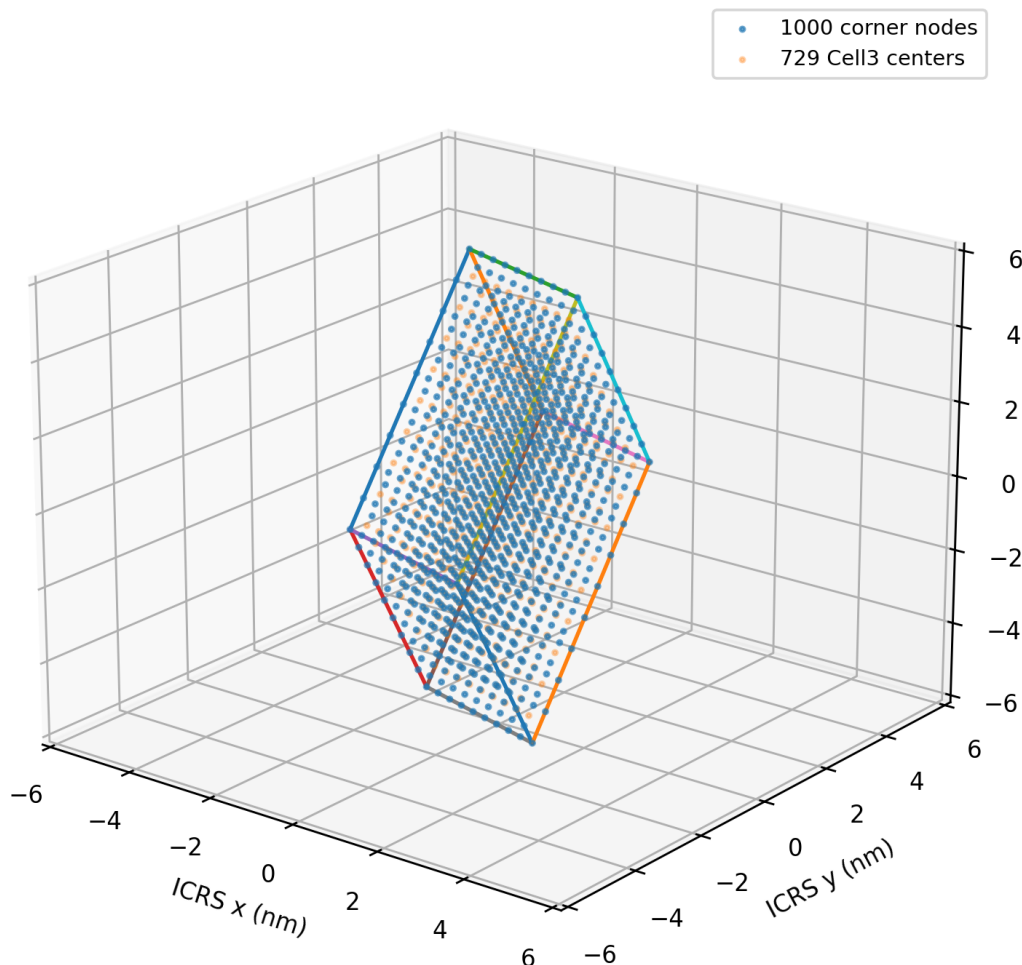


Figure 3.1. Dimensionally correct Lucas Supergroup in ICRS coordinates. The thin cell lattice is a 9 x 9 x 9 tiling; the coarse organization is a 3 x 3 x 3 tiling of Lucas Groups.

3.2 Physical dimensions

Measure	Lucas Supergroup value
Outer edge a	3.007617111531995 nm
Outer edge b	5.277698013700268 nm
Outer edge c	7.130877885384167 nm
Outer face ab	15.784402402299214 nm ²
Outer face ac	21.149605099006632 nm ²
Outer face bc	28.982572669413482 nm ²
Volume	85.959956203661847 nm ³
Opposite-face span a	2.965918767259021 nm
Opposite-face span b	4.064376417491562 nm
Opposite-face span c	5.445879673666999 nm
Body diagonal	12.092882595083326 nm

These absolute nanometre values use the adopted locked realization. Every dimensionless relationship in this report is invariant under common positive rescaling.

3.3 Exact census

Object	Count
Lucas Groups	27
Cell3 volumes	729
Unrefined corner nodes	1000
Half-step nodes per axis	19
Merged half-step nodes	6859
Constant-lambda sheets	55
Central lambda=0 sheet nodes	271
Antipodal half-step pairs	3429
Internal Cell3 interfaces	1944
Exposed primitive faces	486
Cell-center graph cycle rank	1216
Strictly interior Cell3 volumes	343
Strictly interior half-step states	4913

g=2 merged half-step lattice: 6,859 nodes on 55 lambda sheets

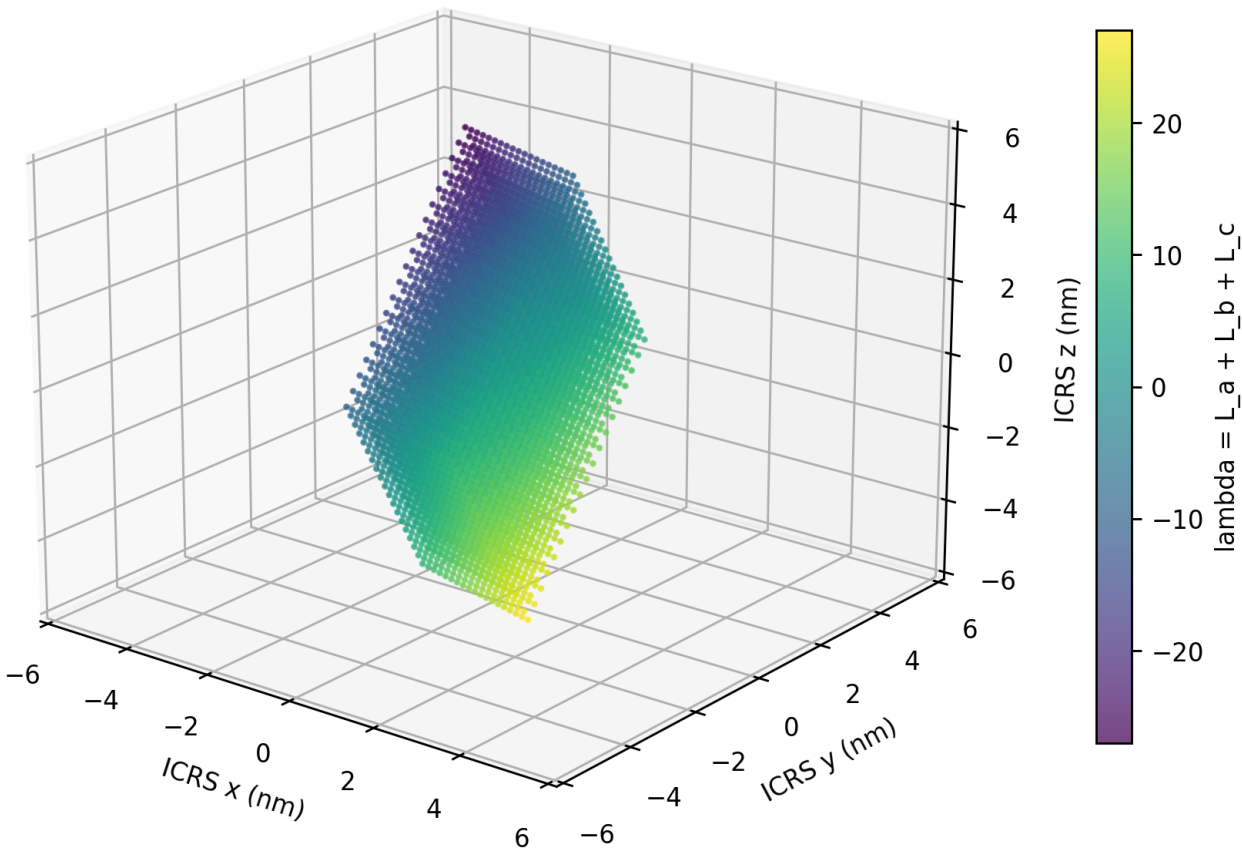


Figure 3.2. Full 19 x 19 x 19 half-step lattice of the Lucas Supergroup. The 6,859 plotted positions include all barycenters of all cellular dimensions.

4. The Half-Step Lattice Is the Complete Cellular Census

4.1 Barycentric correspondence

Along each primitive axis, ten integer-parity positions represent cell boundaries and nine opposite-parity positions represent Cell3 centers. A three-dimensional cellular object of dimension k has k center coordinates and $3-k$ boundary coordinates. Its exact count is

$$f_k = C(3, k) 9^k 10^{(3-k)}.$$

Dimension	Cellular object	Formula	Count
0	Corner vertices	10^3	1000
1	Edge centers	$3 \times 9 \times 10^2$	2700
2	Face centers	$3 \times 9^2 \times 10$	2430
3	Cell3 centers	9^3	729
All	Complete barycentric state set	$(10+9)^3$	6859

Every half-step position is therefore the barycenter of exactly one vertex, edge, face, or three-dimensional Cell3 volume. No half-step state is geometrically untyped.

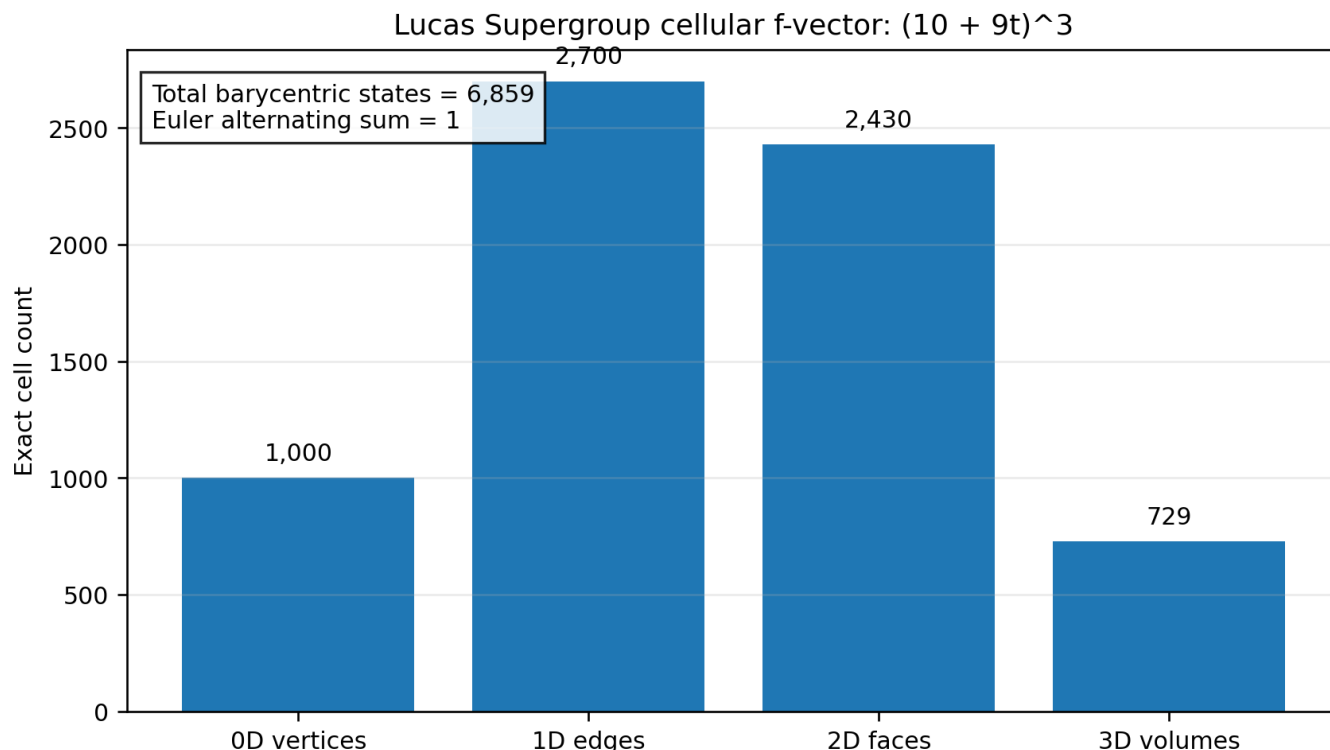


Figure 4.1. Complete cellular f-vector. The coefficients of $(10+9t)^3$ enumerate all zero-, one-, two-, and three-dimensional cells.

4.2 Generating polynomial and topology

$$\begin{aligned}
 F(t) &= (10+9t)^3 \\
 &= 1000 + 2700t + 2430t^2 + 729t^3 \\
 F(1) &= 6859 \\
 F(-1) &= 1000 - 2700 + 2430 - 729 = 1.
 \end{aligned}$$

The evaluation at $t=1$ counts all barycentric states. The evaluation at $t=-1$ is the alternating cellular sum and equals the Euler characteristic of a filled three-dimensional ball. The Supergroup half-step lattice therefore contains a complete cellular topology, not merely a dense set of sample points.

5. Polarity as Cellular Dimension Grading

Let $L=(L_a, L_b, L_c)$ be a half-step integer address with each component from -9 through +9. Boundary and center positions alternate along every axis. The parity $\chi(L)=(-1)^{(L_a+L_b+L_c)}$ therefore changes sign whenever the cellular dimension changes by one. Up to a global convention, $\chi(L)=(-1)^k$ for the barycenter of a k -dimensional cell.

$$\text{Sum}_L \chi(L) = +/- (f_0 - f_1 + f_2 - f_3) = +/- 1.$$

The persistent one-unit checkerboard imbalance is therefore topological. It is the Euler characteristic of the filled Lucas Supergroup. Local polarity and global topology are two readings of the same cellular grading.

Exact result: K-Field checkerboard polarity on the full half-step lattice is the alternating cellular dimension grading.

6. Interior and Boundary Complexes

6.1 Strict interior

Removing all cells incident to the outer boundary leaves eight interior boundary-coordinate positions and nine center-coordinate positions per axis. The interior polynomial is

$$\begin{aligned} F_{\text{int}}(t) &= (8+9t)^3 \\ &= 512 + 1728t + 1944t^2 + 729t^3. \end{aligned}$$

Dimension	Interior count
Vertices	512
Edges	1728
Faces	1944
Cell3 volumes	729
Total interior barycentric states	4913
Alternating sum	-1

The alternating interior sum is -1, the compact-support Euler characteristic of an open three-dimensional ball.

6.2 Outer boundary

$$\begin{aligned} F_{\text{boundary}}(t) &= F(t) - F_{\text{int}}(t) \\ &= 488 + 972t + 486t^2. \end{aligned}$$

Dimension	Boundary count
Vertices	488
Edges	972
Faces	486
Total boundary barycentric states	1946
Euler characteristic	2

The boundary Euler characteristic equals two, identifying the outer cellular surface as a topological two-sphere.

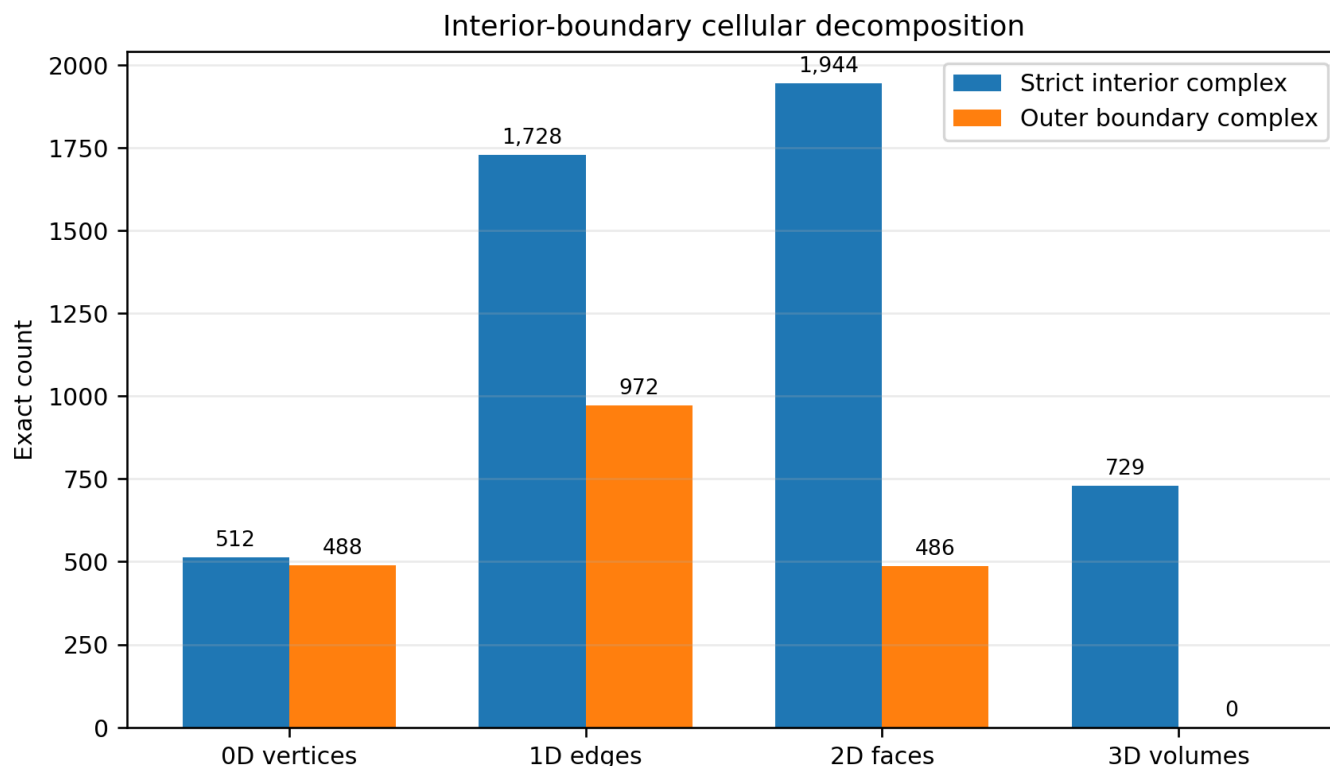


Figure 6.1. Exact decomposition of every cellular dimension into strict-interior and outer-boundary populations.

7. Independent Interface-Boundary Selection of 9B

For a general $m \times m \times m$ block, the internal Cell3 interface count and half-step boundary-state count are

$$I(m) = 3m^2(m-1)$$

$$N_{\text{boundary}}(m) = (2m+1)^3 - (2m-1)^3 = 24m^2 + 2.$$

Demanding that the non-topological part of the boundary census equal the internal interface count gives

$$N_{\text{boundary}}(m) - 2 = I(m)$$

$$24m^2 = 3m^2(m-1)$$

$$m=9.$$

Only the Lucas Supergroup satisfies

$$1946 = 1944 + 2$$

$$\text{boundary barycentric states} = \text{internal interfaces} + \chi(S^2).$$

This derivation does not use the Mass-Bridge face ratios. The projective Mass Bridge and cellular interface-boundary topology independently select the same 9B hierarchy level.

8. Consecutive Cubic Ladders

At the unrefined Cell3 level, the Supergroup produces four consecutive cubes:

$$7^3 = 343 \text{ interior Cell3 volumes}$$

$$8^3 = 512 \text{ interior corner nodes}$$

$$9^3 = 729 \text{ total Cell3 volumes}$$

$$10^3 = 1000 \text{ total corner nodes.}$$

The 343 interior Cell3 volumes equal the complete 7^3 half-step node count of one Lucas Group. The interior of the Supergroup therefore carries a full Lucas Group state count as a volume population.

After exact midpoint refinement, the corresponding ladder is

```
16^3=4096 interior half-cells
17^3=4913 interior half-step nodes
18^3=5832 total half-cells
19^3=6859 total half-step nodes.
```

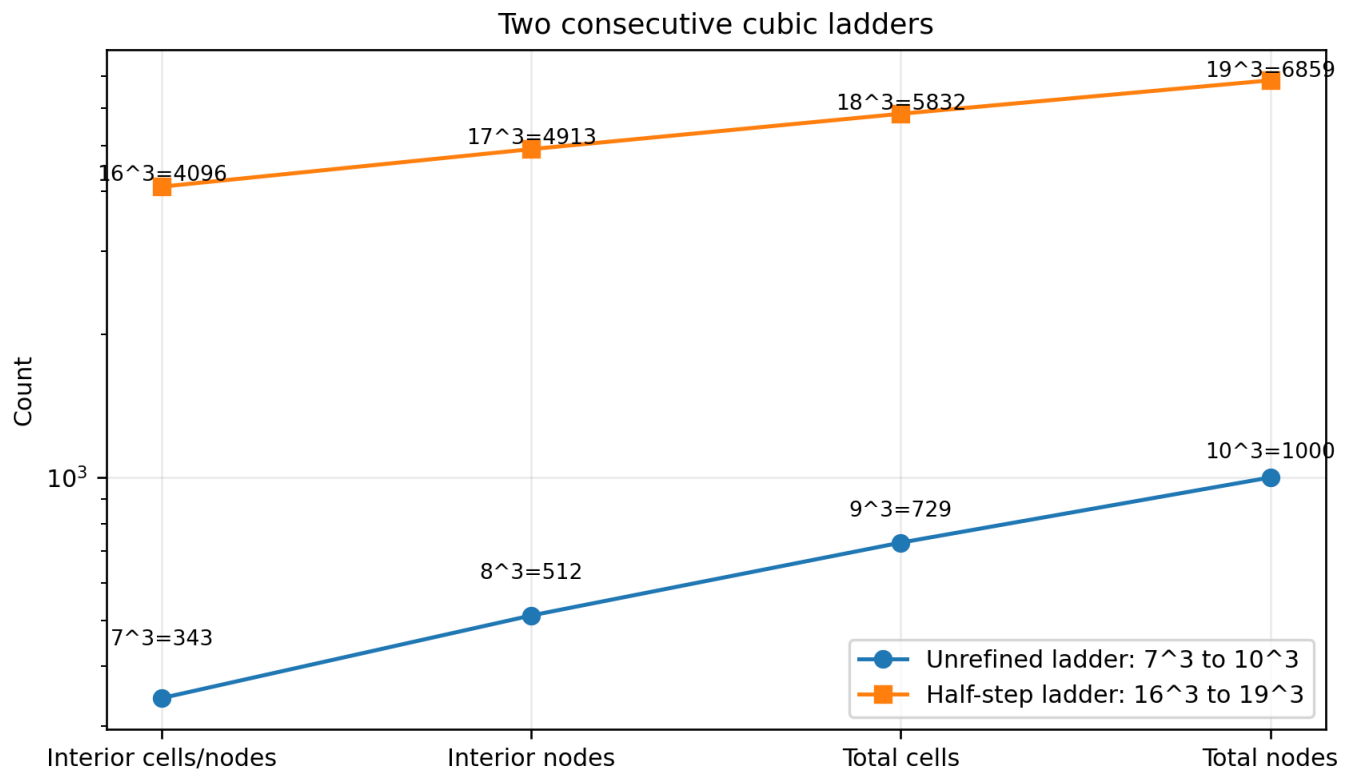


Figure 8.1. The Lucas Supergroup contains two consecutive cubic ladders, one at the unrefined level and one after exact dyadic refinement.

9. Projective Curvature of the Cellular f-Vector

The successive dimensional population ratios are

```
f_1/f_0 = 27/10
f_2/f_1 = 9/10
f_3/f_2 = 3/10.
```

Each ratio is one-third of the preceding ratio. This exact ternary curvature is inherited from three independent primitive directions and the 10-to-9 boundary-center relation along each axis.

```
f_1^2 = 3 f_0 f_2
f_2^2 = 3 f_1 f_3
f_1 f_2 = 9 f_0 f_3.
```

The generating polynomial has the triple root $t=-10/9$. The scalar $10/9$ is the node-to-cell ratio along one primitive direction; the triple root records its independent replication through three dimensions.

10. Finite Geometric Derivatives

Let $V(m)=m^3$ be the accumulated Cell3 volume count. At $m=9$, exact finite differences generate several Supergroup counts.

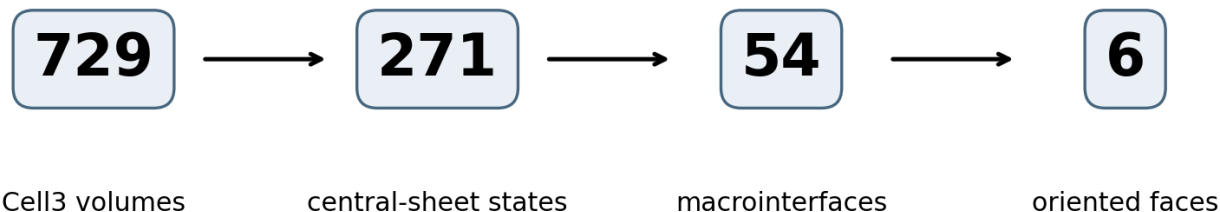
```
V(9)=729
V(10)-V(9)=271
V(10)-2V(9)+V(8)=54
Delta^3 V=6.
```

The first difference is the central-sheet population. The centered second difference is the number of macrointerfaces between 27 Lucas Groups. The third difference is the six oriented faces of a parallelepiped. The resulting derivative ladder is

729 -> 271 -> 54 -> 6.

The total sheet count is $55=54+1$, so it combines the second geometric derivative with the bulk Euler unit.

Finite geometric derivative ladder of $V(m)=m^3$ at $m=9$



$$729 \rightarrow (10^3 - 9^3) = 271 \rightarrow (10^3 - 2 \cdot 9^3 + 8^3) = 54 \rightarrow \Delta^3 m^3 = 6$$

Figure 10.1. Exact derivative ladder of the cubic accumulation law. Each derivative acquires a distinct geometric meaning within the Lucas Supergroup.

11. Exact Interface Decomposition

The 1,944 internal Cell3 interfaces separate into intragroup and intergroup contributions.

$27 \times 54 = 1458$ intragroup interfaces
 $54 \times 9 = 486$ intergroup Cell3-face interfaces
 $1944 = 1458 + 486$.

The outer surface also contains 486 Cell3 faces. Therefore

intragroup : intergroup : external = 1458 : 486 : 486 = 3 : 1 : 1.

At the Lucas-Group macrolevel, 54 intergroup macrointerfaces exactly balance 54 external macrofaces. The local 54-to-54 incidence balance survives at the Supergroup scale.

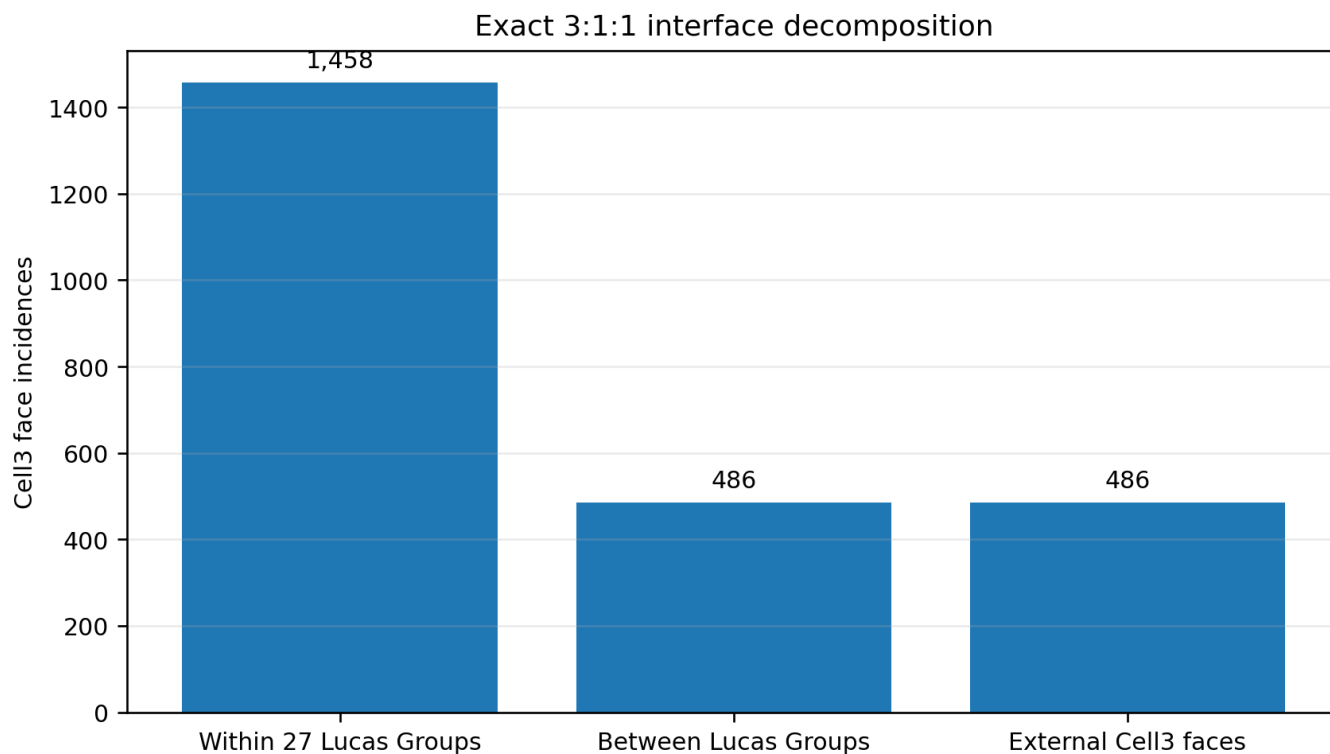


Figure 11.1. Exact 3:1:1 decomposition of all internal and exposed Cell3-face incidences.

12. Two-Level Balanced-Ternary Address

Every Cell3 center has a unique hierarchical address

$$\begin{aligned} x &= 3G + c \\ G &\text{ in } \{-1, 0, 1\}^3 \\ c &\text{ in } \{-1, 0, 1\}^3. \end{aligned}$$

The coarse triple G selects one of 27 Lucas Groups. The local triple c selects one of 27 Cell3 positions within that group. The 729 centers are therefore exact six-digit balanced-ternary addresses, not an undifferentiated count.

$$\begin{aligned} \chi(x) &= (-1)^{(x_a + x_b + x_c)} = \chi(G) \chi(c) \\ \lambda_x &= 3 \lambda_G + \lambda_c. \end{aligned}$$

Because three is odd, local and collective polarity multiply. Sheet coordinates combine linearly, while multiple coarse-local combinations can occupy the same total sheet. This produces exact interlevel sheet mixing.

Two-level balanced-ternary address of every Cell3 center

$$\boxed{\begin{array}{l} G \text{ in } \{-1,0,1\}^3 \\ 27 \text{ Lucas Groups} \end{array}} \longrightarrow \mathbf{x = 3G + c} \longleftarrow \boxed{\begin{array}{l} c \text{ in } \{-1,0,1\}^3 \\ 27 \text{ Cell3 positions} \end{array}}$$

729 unique six-digit ternary addresses; parity factorizes as $\chi(x) = \chi(G) \chi(c)$

Phase labels form Z_9^3 with $q=r+3s$; ternary carry couples the two Z_3^3 levels.

Figure 12.1. Two-level balanced-ternary addressing of all 729 Cell3 centers and the corresponding coupled phase representation.

13. Nontrivial Ternary Phase Extension

With periodic boundary conditions, Supergroup translation and Bloch labels form Z_9^3 . Every phase index may be written $q=r+3s$ with r,s in Z_3^3 . This gives 27 coarse residue sectors and 27 fine sectors within each coarse sector.

$$0 \rightarrow Z_3^3 \rightarrow Z_9^3 \rightarrow Z_3^3 \rightarrow 0.$$

The extension does not split as $Z_3^3 \times Z_3^3$. Ternary carry couples the local and collective digits. Thus the Lucas Supergroup contains two Lucas-scale phase levels, but their phases are not independent. This cross-scale carry is an intrinsic emergent property of the 9B block.

14. Exact Laminar Population Law

The half-step coordinates range from -9 through +9 and the laminar coordinate is $\lambda = L_a + L_b + L_c$. Hence $-27 \leq \lambda \leq 27$ and there are 55 sheets. Their populations are coefficients of

$$(x^{-9} + x^{-8} + \dots + x^8 + x^9)^3.$$

The exact piecewise law is

$$\begin{aligned} N(\lambda) &= 271 - \lambda^2, \text{ for } |\lambda| \leq 9 \\ N(\lambda) &= C(29 - |\lambda|, 2), \text{ for } 9 \leq |\lambda| \leq 27. \end{aligned}$$

The inner nineteen sheets lie on an exact discrete parabola. The outer tails are triangular-number sequences. The complete distribution sums to 6,859 and is symmetric under λ reversal.

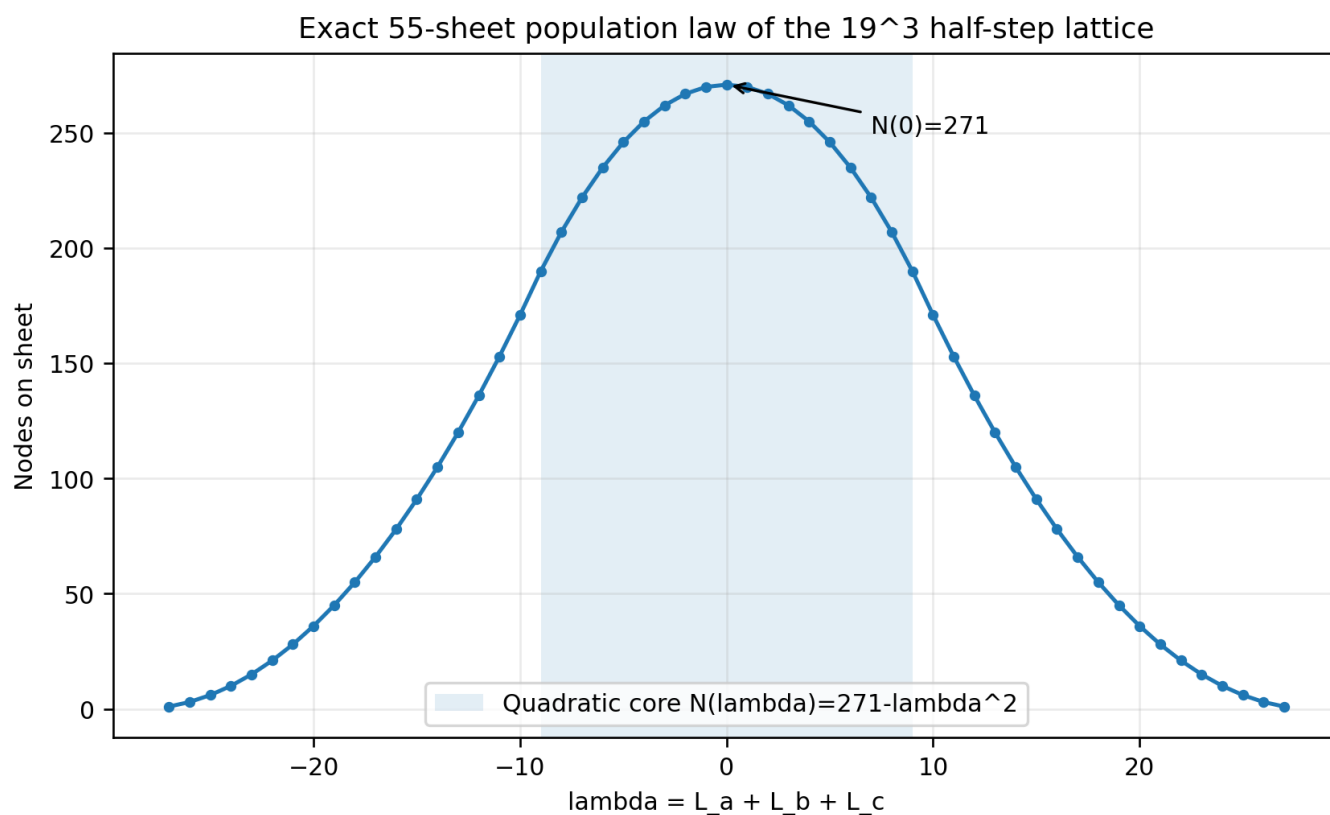


Figure 14.1. Exact population of all 55 laminar sheets. The central region is quadratic and the outer regions are triangular tails.

15. Super-Emily Spectral Hierarchy

For the nearest-neighbor Cartesian product graph $P_n \square P_n \square P_n$, direct diagonalization was performed at the hierarchy-relevant odd sizes $n=3, 7, 9$, and 19 .

Graph role	n	Vertices n^3	Sheets $3n-2$	Computed nullity	Rank
Cell3 half-step graph	3	27	7	7	20
Lucas Group half-step graph	7	343	19	19	324
Supergroup center graph	9	729	25	25	704
Supergroup half-step graph	19	6859	55	55	6804

At all four hierarchy-relevant sizes, the adjacency zero-mode dimension equals the number of constant-sum laminar sheets. For the full Supergroup half-step graph, the nullity is exactly 55.

Super-Emily result: the 55 geometric sheets are simultaneously 55 independent zero-adjacency modes.

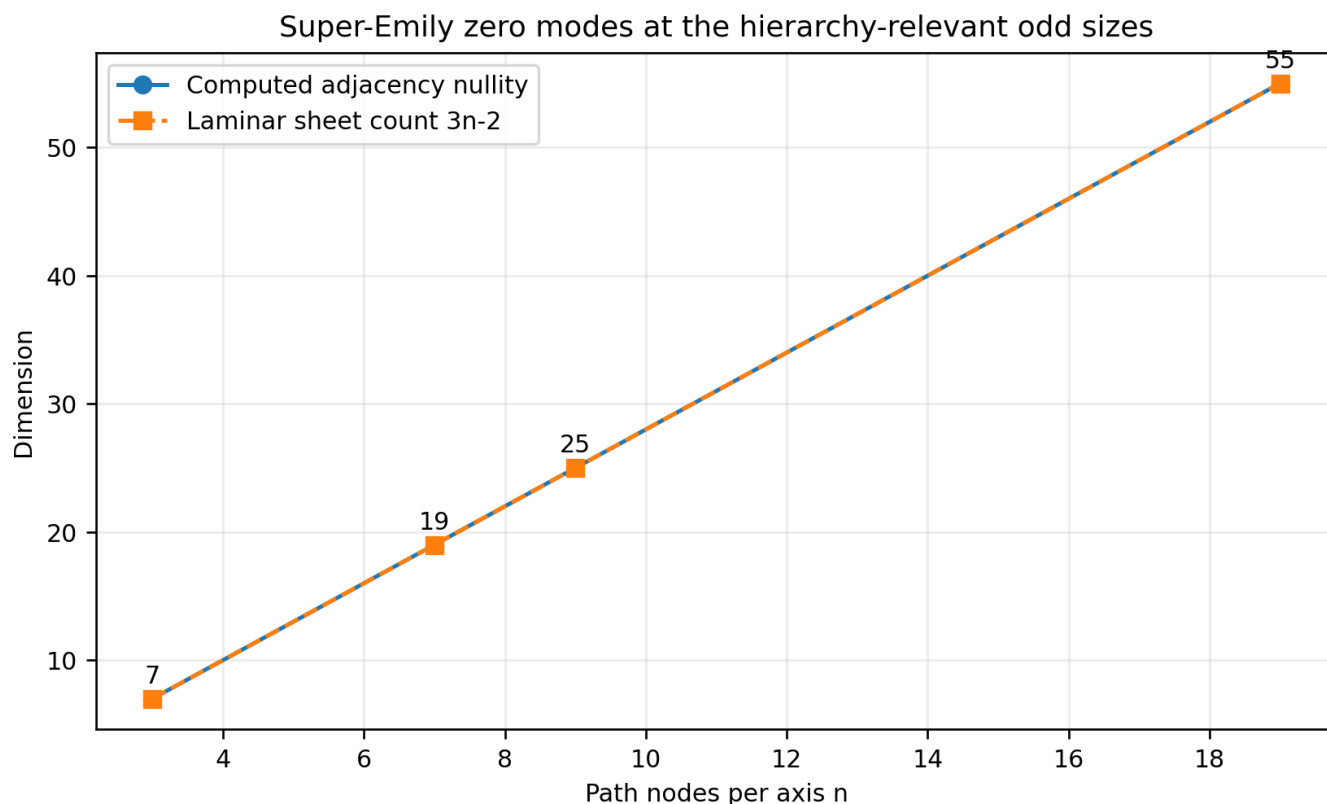


Figure 15.1. Computed zero-mode dimensions and laminar sheet counts for the four hierarchy-relevant path-product graphs.

16. Sheet-to-Volume and Eigen-to-Volume Recursion

At aggregation depth g , the laminar sheet count is $S_g = 6(3^g) + 1$ and the next level has half-step node count $Q_{(g+1)} = (2 \cdot 3^{(g+1)} + 1)^3$. These satisfy

$$S_g^3 = Q_{(g+1)}.$$

$$7^3 = 343$$

$$19^3 = 6859$$

$$55^3 = 166375.$$

Because the adjacency nullity equals the sheet count at the hierarchy-relevant levels, a one-dimensional zero-mode count at one level becomes the complete three-dimensional state count at the next. This is an exact eigen-to-volume recursion.

17. Cycle-Space Dimension and Primal-Dual Equality

The Cell3-center graph contains $V_c = 729$ vertices and $E_c = 1,944$ nearest-neighbor edges. Its connected graph cycle rank is

$$\begin{aligned} \beta_1(\text{graph}) &= E_c - V_c + 1 = 1944 - 729 + 1 = 1216 \\ 1216 &= 8^2 \times 19. \end{aligned}$$

The strict-interior corner skeleton has 512 vertices and 1,728 edges, giving

$$1728 - 512 = 1216.$$

The center-graph cycle rank therefore equals the interior corner-edge surplus. This is an exact primal-dual equality between circulation capacity in the center graph and edge excess in the interior corner skeleton. The filled complex is still simply connected; two-dimensional faces close these graph cycles.

18. Chebyshev Shell Hierarchy

Define shell radius $r = \max(|L_a|, |L_b|, |L_c|)$. The shell and antipodal-pair counts are

$$N_r = (2r+1)^3 - (2r-1)^3 = 24r^2 + 2$$

$$P_r = 12r^2 + 1.$$

Radius	Hierarchy object	Shell nodes	Antipodal pairs	New primitive pairs	Inherited pairs
1	Cell3 boundary	26	13	13	0
3	Lucas Group boundary	218	109	96	13
9	Lucas Supergroup boundary	1946	973	864	109

The local 13 antipodal channels are the first member of the sequence 13, 109, 973. Each larger shell contains inherited lower-level directions plus directions primitive to the new radius.

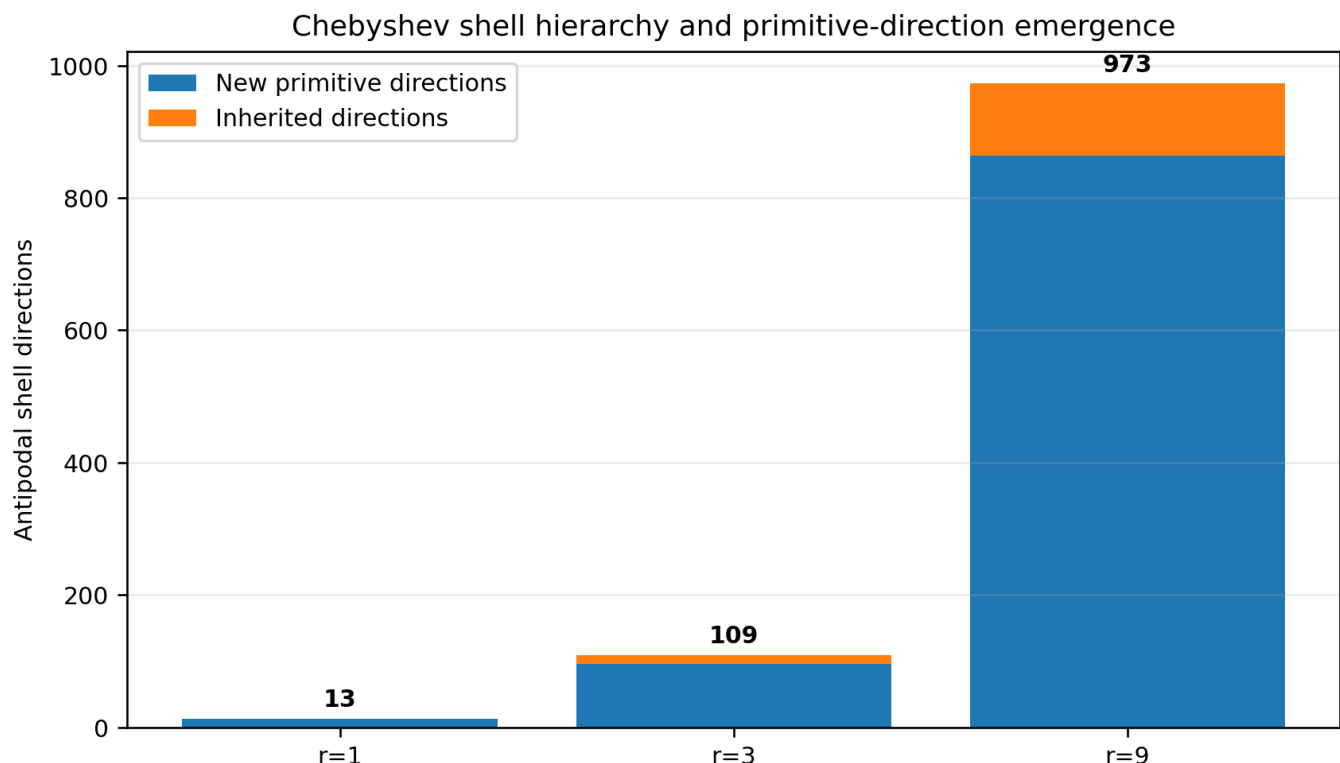


Figure 18.1. Total, inherited, and newly primitive antipodal directions at radii 1, 3, and 9.

19. Primitive-Direction Recursion and the Emergence of 864

A shell direction is primitive when $\gcd(|L_a|, |L_b|, |L_c|) = 1$. For radius r , the directed primitive count is $24 J_2(r)$, where J_2 is the second Jordan totient. Antipodal pairing gives $12 J_2(r)$.

$$J_2(9) = 9^2(1 - 1/3^2) = 72$$

$$12 J_2(9) = 864.$$

The Supergroup outer shell therefore contains exactly 864 new primitive antipodal directions and 109 inherited directions, with

$$973 = 864 + 109.$$

The inherited 109 directions are exactly the complete antipodal shell of the Lucas Group at radius three. The new primitive counts grow by a factor of nine per ternary aggregation: 96, 864, 7,776, and so on.

19.1 Physical correspondence

The independently measured electron-to-shielded-helion magnetic-moment ratio in the source comparison is 864.058255000. The exact Supergroup primitive-direction count is 864. The signed relative residual is -6.742022272566e-05, or 67.420223 ppm.

864 = primitive antipodal directions on the radius-9 Supergroup shell.

This geometric origin is more specific than the equivalent factorization $2^5 \times 3^3$. The count is exact and uniquely associated with the Mass-Bridge-selected object. The physical mapping remains a candidate until a mechanism links the primitive reciprocal-direction census to the magnetic-moment ratio and predicts additional quantities without object-specific fitting.

20. Primitive q-Spectrum and Compatibility Space

For every primitive antipodal address s , define $q(s)=s^T R s$. The 864 primitive directions generate 864 numerically distinct q -values for the generic locked reciprocal metric.

$$q(s)=s_a^2 R_{aa}+s_b^2 R_{bb}+s_c^2 R_{cc}+2s_as_b R_{ab}+2s_as_c R_{ac}+2s_bs_c R_{bc}.$$

A symmetric 3 x 3 metric has six independent entries. The primitive shell therefore supplies

$$864 \text{ observations} - 6 \text{ metric coordinates} = 858 \text{ compatibility dimensions}.$$

This generalizes the local Lucas result $13-6=7$. At the Supergroup level, the metric is massively overdetermined by the primitive shell.

Quantity	Value
Primitive antipodal directions	864
Observation matrix rank	6
Compatibility dimension	858
Distinct q values at 1e-10	864
Minimum q	130.593523477839 nm ⁻²
Maximum q	1744.361292059535 nm ⁻²
Mean q	757.916780244175 nm ⁻²
Population standard deviation	325.034198848754 nm ⁻²

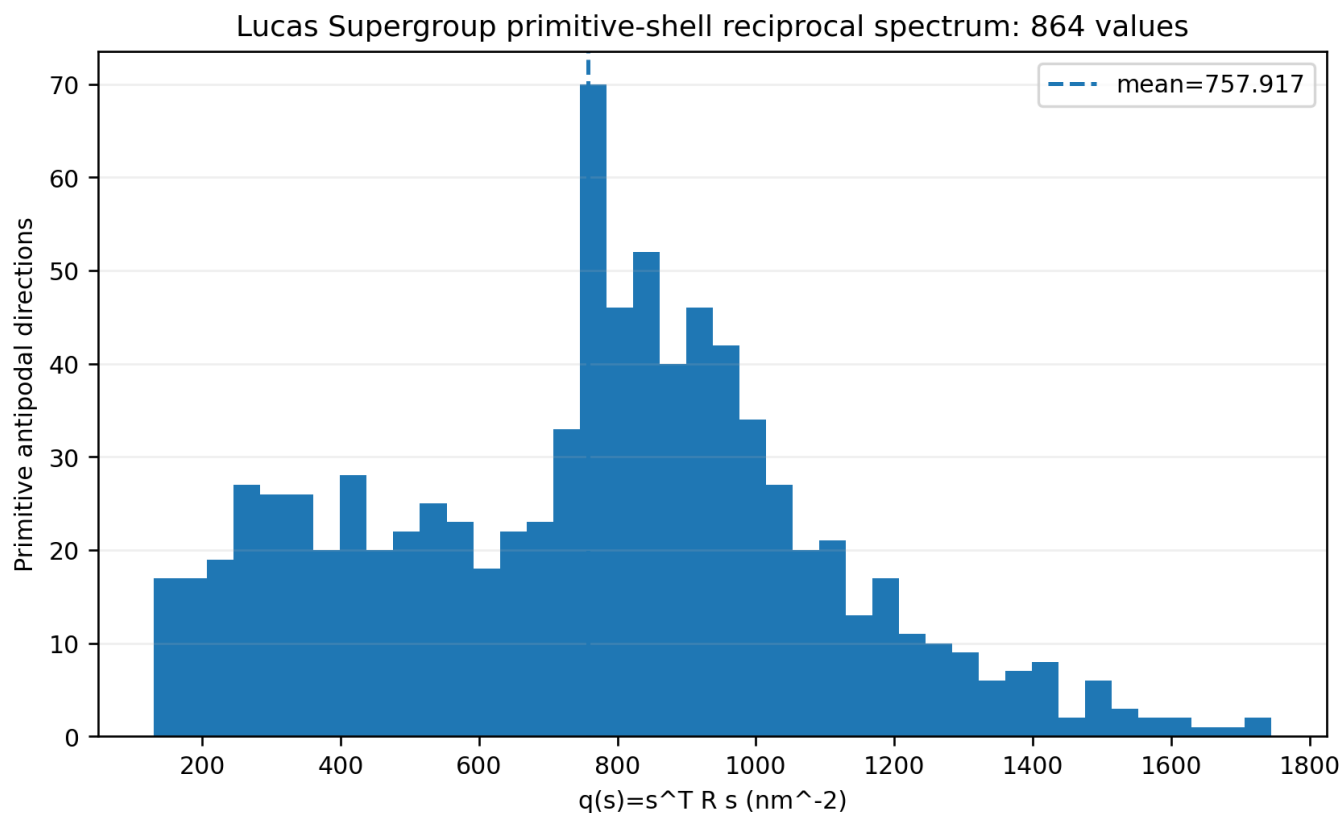


Figure 20.1. Distribution of the 864 primitive reciprocal quadratic values $q(s)=s^T R s$.

21. Hidden 1 + 2 + 3 Metric-Observation Eigensystem

Each q observation is linear in the six independent entries of R . The 864×6 observation matrix has full rank six. Its Gram operator separates exactly into three invariant sectors.

Sector	Dimension	Squared singular value	Interpretation
Common trace	1	5,584,752	Uniform reciprocal scale
Diagonal traceless A2	2	1,175,976	Two-dimensional diagonal shape sector
Off-diagonal shear	3	5,878,368	Three independent reciprocal couplings

Symmetric metric space = 1 trace scalar + 2 A2 shape coordinates + 3 shear coordinates.

The primitive shell therefore resolves the repeated one-plus-two shape decomposition and extends it by an independent three-dimensional shear triplet. This is the complete six-dimensional eigen-like organization of the reciprocal metric observation problem.

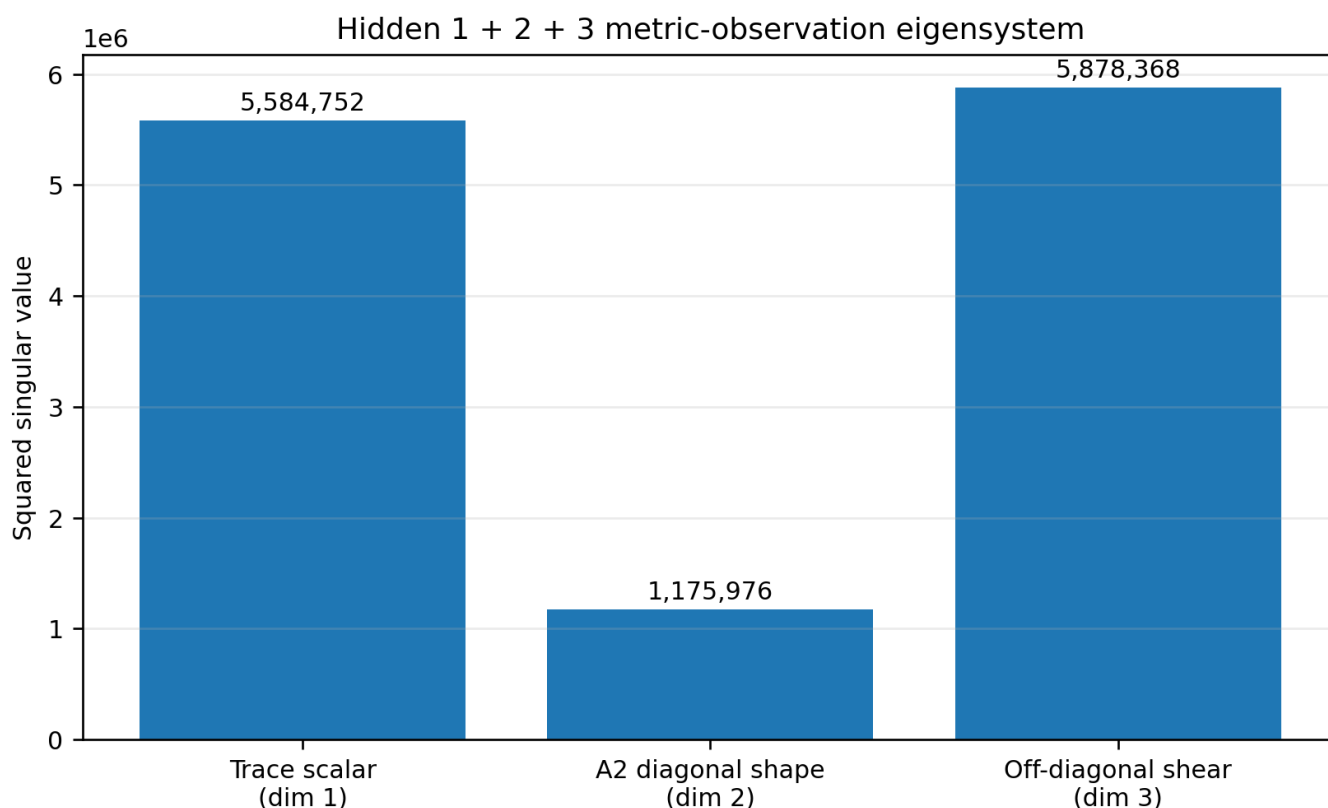


Figure 21.1. Exact sector decomposition of the primitive-shell observation Gram operator.

22. Reciprocal Principal Dimensions and Integer Eigenaddresses

Diagonalizing the locked reciprocal metric yields three orthogonal reciprocal principal modes. Their inverse eigenvalues give the direct principal squared lengths. Multiplying the direct principal lengths by nine gives the corresponding orthogonal Supergroup spans.

Mode	ρ (nm ⁻²)	Cell3 principal length (nm)	9B span (nm)	Best integer eigenaddress	Angular error	Rayleigh relative error
soft	1.222872851445	0.904293373154	8.138640358390	(-1, -6, -9)	2.046274 deg	0.708467%
intermediate	6.333432714032	0.397356589534	3.576209305803	(-1, 9, -6)	2.460792 deg	-0.049579%
hard	9.286289551060	0.328154896945	2.953394072508	(-9, -1, 1)	2.706425 deg	-0.130524%

The direct-metric condition number is 7.593830822301. The integer addresses (1,6,9), (1,-9,6), and (9,1,-1), up to antipodal sign, are high-accuracy discrete representatives of the soft, intermediate, and hard reciprocal principal axes.

Reciprocal metric ellipsoid and exact integer eigenaddresses

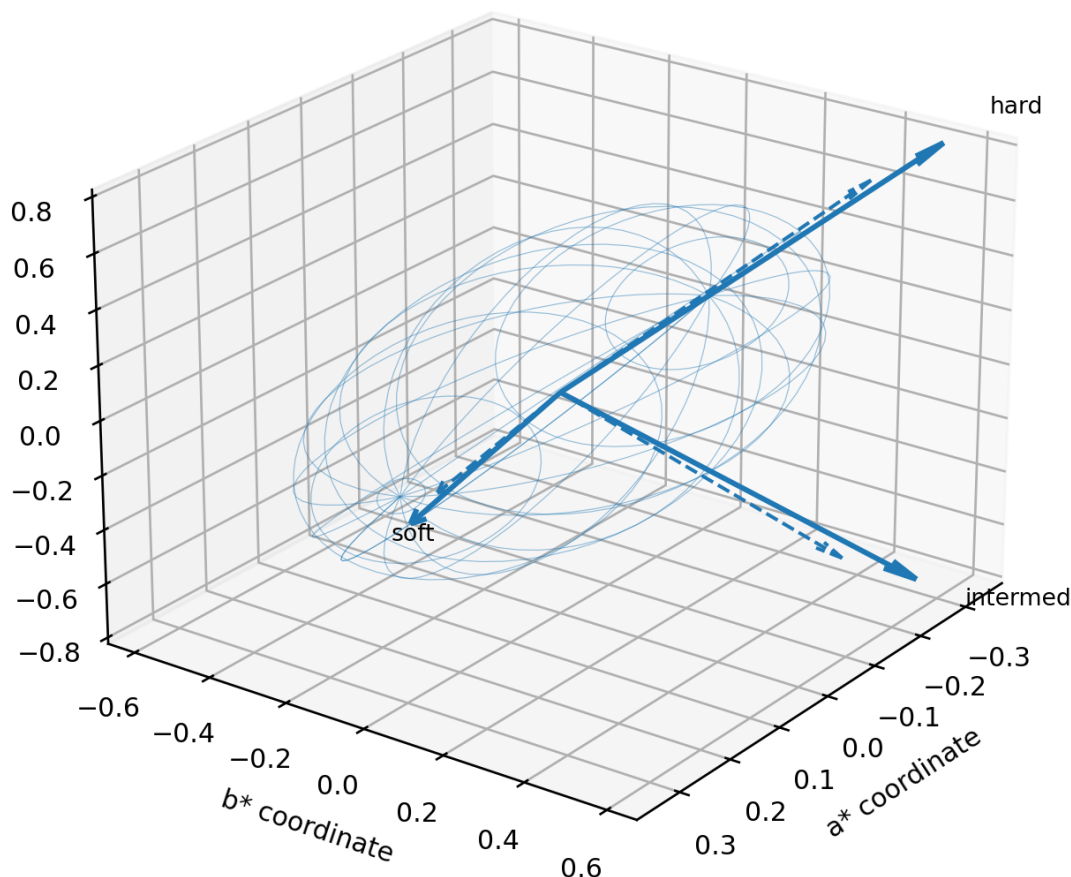


Figure 22.1. Reciprocal metric ellipsoid. Solid rays show continuous principal axes; dashed rays show the nearest exact primitive integer eigenaddresses available on the Supergroup shell.

23. Reciprocal Mean and Anisotropy Decomposition

The primitive shell is combinatorially balanced. Summing over its 864 antipodal representatives gives equal coordinate-square sums and vanishing cross sums.

$$\begin{aligned} \text{Sum } s_a^2 &= \text{Sum } s_b^2 = \text{Sum } s_c^2 = 38880 \\ \text{Sum } s_{as_b} &= \text{Sum } s_{as_c} = \text{Sum } s_{bs_c} = 0. \end{aligned}$$

Therefore

$$\langle q \rangle = 45 \text{ trace}(R) = 757.916780244175 \text{ nm}^{-2}.$$

The address shell supplies an isotropically balanced measurement design, while the locked oblique metric splits that design into a fully anisotropic physical spectrum. Combinatorial symmetry and metric anisotropy are cleanly separated.

24. Direct and Reciprocal Symmetry Separation

Before metric weighting, the primitive address shell has signed-permutation symmetry. Directions related by axis permutations and sign changes belong to the same combinatorial class. The locked metric has unequal diagonals and nonzero couplings, so it splits those classes.

address symmetry -> metric splitting -> physical q-spectrum.

Under the locked metric, all 864 primitive antipodal q-values are distinct at the stated numerical precision. Antipodal inversion is already quotiented out. The Lucas Supergroup therefore functions as a finite directional spectrometer for the complete reciprocal metric.

25. Object-to-Field Transition

The Lucas Supergroup contains 343 strictly interior Cell3 volumes, or 47.050754 percent of its volume cells. It contains 4,913 strictly interior half-step states, or 71.628517 percent of its complete cellular-state lattice.

The geometric volume population is still close to its boundary transition, while the cellular-state population is already strongly bulk dominated. Field-like behavior therefore emerges first in the complete cellular-state complex rather than in the raw count of three-dimensional cells.

Cell3 metric -> Lucas collective neighborhood -> Supergroup complete cellular field complex -> larger bulk continuum.

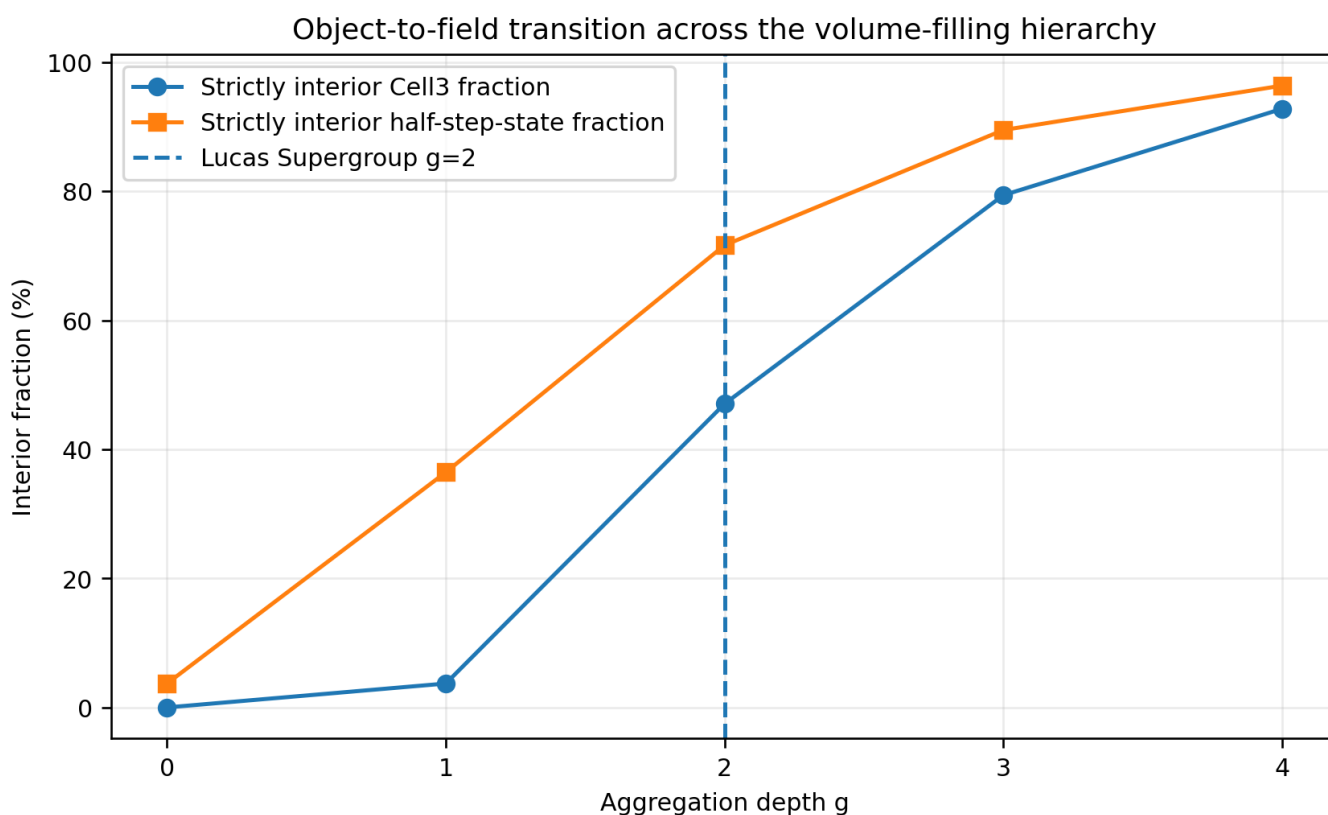


Figure 25.1. Interior fractions across aggregation depth. At $g=2$, half-step cellular states are already bulk dominated while complete Cell3 volumes remain near the transition.

26. Synthesis of Emergent Properties and Hidden Dimensions

Emergent object	Exact value	Meaning
Bulk Euler mode	1	Euler characteristic of the filled cellular complex
Boundary Euler modes	2	Euler characteristic of the outer two-sphere
Center-graph zero modes	25	Nullity of P_9 square P_9 square P_9
Half-step zero modes	55	Nullity of P_{19} square P_{19} square P_{19} and laminar sheet count
Graph circulation dimensions	1216	Independent nearest-neighbor center-graph cycles before face closure
Primitive directions	864	New primitive antipodal radius-9 shell directions
Primitive q compatibility dimensions	858	864 reciprocal observations minus six metric coordinates
Metric sector decomposition	$1 + 2 + 3$	Trace scalar, A2 shape doublet, shear triplet

These dimensions are not independent numerology. Each arises from an explicitly defined geometry, cellular chain, graph, shell, or matrix. Together they show that the Lucas Supergroup is simultaneously a space-filling object, a complete finite cell complex, a parity-graded topology, a laminar resonator, a graph-spectral system, and a reciprocal metric measurement frame.

27. Conclusions and Forward Research Program

The Lucas Supergroup is not merely a larger Cell3 block. It is the first level at which the Mass-Bridge cardinalities, complete cellular topology, checkerboard polarity, ternary hierarchy, laminar spectrum, graph nullspaces, and reciprocal primitive directions close in one finite object.

The strongest new exact result is the identity between half-step positions and every cellular barycenter. This converts the 19^3 lattice into a complete chain-complex census. The alternating polarity sum becomes the Euler characteristic. The boundary carries Euler characteristic two. The unique equality between boundary-state count and internal-interface count plus two independently selects $m=9$.

The strongest new spectral result is the 55-dimensional Super-Emily nullspace, equal to the 55 laminar sheets. The strongest new reciprocal result is the 864-direction primitive shell and its 858-dimensional compatibility space. The observation matrix resolves the symmetric metric into one scale coordinate, two A2 shape coordinates, and three shear coordinates.

The next strict program is to construct metric-weighted cellular boundary operators and the complete k-form Hodge spectra of the 1000-vertex, 2700-edge, 2430-face, 729-volume complex. This will determine how the 1,216 center-graph circulation modes are closed by faces, how the 55 laminar zero modes distribute among scalar, vector, and tensor sectors, and whether the 864-direction primitive reciprocal shell predicts measured field or particle spectra under one global dimensional scale and no object-specific coefficients.

Final conclusion: The Lucas Supergroup is the first complete cellular, topological, spectral, and reciprocal closure of Cell3.

References and Source Record

- Krytronx. K-Field Mass-Bridge Locked Projective Ratio System. JSON epoch 1785176592. SHA-256 a5ba3ddf21a20e4528c19609a3061fcca60f53f76224f101c9779c24809d7ea.
- Krytronx. K-Field Mass Bridge Drives Volume-Filling Hierarchy. PDF and JSON epoch 1785204375. JSON SHA-256 811c98f49447534c27d14fd77c779d756e4acc73bf44bfbf93ae7bebf81e79b2.
- Krytronx. K-Field Fractional Recursive Quanta - Lucas Group. Machine-readable hierarchy source epoch 1784239728.
- Krytronx. K-Field Lucas Group Standing-Wave Cavity Analysis. Finite translation sectors, half-step nodes, and laminar sheets.
- Krytronx. Commutation Lucas-q / Cell3 Reversible. Local 13-to-6 q compatibility and seven-dimensional Emily nullspace.
- CODATA 2022 recommended constants as recorded in the machine-readable source comparison.

Appendix A. Complete Locked Basis and Metrics

Cartesian row	a	b	c
ICRS x	0.327501242055891	0.035330775843828	-0.010129317626729
ICRS y	-0.054897007884006	0.207319850440192	-0.389787702366260
ICRS z	-0.037487502894730	-0.547401085388508	-0.689734408367268

Direct metric row	a	b	c
0	0.111676057896050	0.020710333280606	0.043937235096220
1	0.020710333280606	0.343877732392787	0.296392758961523
2	0.043937235096220	0.296392758961523	0.627770610077296

Reciprocal metric row	a*	b*	c*
0	9.208025529457549	0.001534802428193	-0.645188035699939
1	0.001534802428193	4.903398312367987	-2.315175584416389
2	-0.645188035699939	-2.315175584416389	2.731171274711695

Appendix B. Complete 27-Node Ledger

Address	lambda	Polarity	Class	ICRS position (nm)
(-1, -1, -1)	-3	-1	corner	-0.176351350136495, 0.118682429905037, 0.637311498325253
(-1, -1, 0)	-2	1	edge center	-0.181416008949859, -0.076211421278093, 0.292444294141619
(-1, -1, 1)	-1	-1	corner	-0.186480667763224, -0.271105272461223, -0.052422910042015
(-1, 0, -1)	-2	1	edge center	-0.158685962214581, 0.222342355125133, 0.363610955630999
(-1, 0, 0)	-1	-1	face center	-0.163750621027945, 0.027448503942003, 0.018743751447365
(-1, 0, 1)	0	1	edge center	-0.168815279841310, -0.167445347241127, -0.326123452736269
(-1, 1, -1)	-1	-1	corner	-0.141020574292667, 0.326002280345229, 0.089910412936745
(-1, 1, 0)	0	1	edge center	-0.146085233106031, 0.131108429162099, -0.254956791246889
(-1, 1, 1)	1	-1	corner	-0.151149891919396, -0.063785422021031, -0.599823995430523
(0, -1, -1)	-2	1	edge center	-0.012600729108550, 0.091233925963034, 0.618567746877888
(0, -1, 0)	-1	-1	face center	-0.017665387921914, -0.103659925220096, 0.273700542694254
(0, -1, 1)	0	1	edge center	-0.022730046735279, -0.298553776403226, -0.071166661489380
(0, 0, -1)	-1	-1	face center	0.005064658813364, 0.194893851183130, 0.344867204183634
(0, 0, 0)	0	1	body center	0.000000000000000, 0.000000000000000, 0.000000000000000
(0, 0, 1)	1	-1	face center	-0.005064658813364, -0.194893851183130, -0.344867204183634
(0, 1, -1)	0	1	edge center	0.022730046735279, 0.298553776403226, 0.071166661489380
(0, 1, 0)	1	-1	face center	0.017665387921914, 0.103659925220096, -0.273700542694254
(0, 1, 1)	2	1	edge center	0.012600729108550, -0.091233925963034, -0.618567746877888
(1, -1, -1)	-1	-1	corner	0.151149891919396, 0.063785422021031, 0.599823995430523
(1, -1, 0)	0	1	edge center	0.146085233106031, -0.131108429162099, 0.254956791246889
(1, -1, 1)	1	-1	corner	0.141020574292667, -0.326002280345229, -0.089910412936745
(1, 0, -1)	0	1	edge center	0.168815279841310, 0.167445347241127, 0.326123452736269
(1, 0, 0)	1	-1	face center	0.163750621027945, -0.027448503942003, -0.018743751447365
(1, 0, 1)	2	1	edge center	0.158685962214581, -0.222342355125133, -0.363610955630999
(1, 1, -1)	1	-1	corner	0.186480667763224, 0.271105272461223, 0.052422910042015
(1, 1, 0)	2	1	edge center	0.181416008949859, 0.076211421278093, -0.292444294141619
(1, 1, 1)	3	-1	corner	0.176351350136495, -0.118682429905037, -0.637311498325253

Appendix C. Six Boundary Planes and Node Membership

Channel	Sign	Primitive coordinate	Unit normal	Area	Spacing	Nine node addresses
a	-1	-0.5	-0.995965190309, 0.083602162507, -0.032619290564	0.35780953 9129	0.329546529 695	(-1, -1, -1); (-1, -1, 0); (-1, -1, 1); (-1, 0, -1); (-1, 0, 0); (-1, 0, 1); (-1, 1, -1); (-1, 1, 0); (-1, 1, 1)
a	1	0.5	-0.995965190309, 0.083602162507, -0.032619290564	0.35780953 9129	0.329546529 695	(1, -1, -1); (1, -1, 0); (1, -1, 1); (1, 0, -1); (1, 0, 0); (1, 0, 1); (1, 1, -1); (1, 1, 0); (1, 1, 1)
b	-1	-0.5	-0.089052594102, -0.866576769291, 0.491033948326	0.26110623 5790	0.451597379 721	(-1, -1, -1); (-1, -1, 0); (-1, -1, 1); (0, -1, -1); (0, -1, 0); (0, -1, 1); (1, -1, -1); (1, -1, 0); (1, -1, 1)
b	1	0.5	-0.089052594102, -0.866576769291, 0.491033948326	0.26110623 5790	0.451597379 721	(-1, 1, -1); (-1, 1, 0); (-1, 1, 1); (0, 1, -1); (0, 1, 0); (0, 1, 1); (1, 1, -1); (1, 1, 0); (1, 1, 1)
c	-1	-0.5	0.194092200808, 0.913177168821, 0.358379234791	0.19486916 5460	0.605097741 519	(-1, -1, -1); (-1, 0, -1); (-1, 1, -1); (0, -1, -1); (0, 0, -1); (0, 1, -1); (1, -1, -1); (1, 0, -1); (1, 1, -1)
c	1	0.5	0.194092200808, 0.913177168821, 0.358379234791	0.19486916 5460	0.605097741 519	(-1, -1, 1); (-1, 0, 1); (-1, 1, 1); (0, -1, 1); (0, 0, 1); (0, 1, 1); (1, -1, 1); (1, 0, 1); (1, 1, 1)

Appendix D. Complete 55-Sheet Population Ledger

lambda	Population	Closed-form branch
-27	1	$C(29- \lambda , 2)$
-26	3	$C(29- \lambda , 2)$
-25	6	$C(29- \lambda , 2)$
-24	10	$C(29- \lambda , 2)$
-23	15	$C(29- \lambda , 2)$
-22	21	$C(29- \lambda , 2)$
-21	28	$C(29- \lambda , 2)$
-20	36	$C(29- \lambda , 2)$
-19	45	$C(29- \lambda , 2)$
-18	55	$C(29- \lambda , 2)$
-17	66	$C(29- \lambda , 2)$
-16	78	$C(29- \lambda , 2)$
-15	91	$C(29- \lambda , 2)$
-14	105	$C(29- \lambda , 2)$
-13	120	$C(29- \lambda , 2)$
-12	136	$C(29- \lambda , 2)$
-11	153	$C(29- \lambda , 2)$
-10	171	$C(29- \lambda , 2)$
-9	190	$271-\lambda^2$
-8	207	$271-\lambda^2$
-7	222	$271-\lambda^2$
-6	235	$271-\lambda^2$
-5	246	$271-\lambda^2$
-4	255	$271-\lambda^2$
-3	262	$271-\lambda^2$
-2	267	$271-\lambda^2$
-1	270	$271-\lambda^2$
0	271	$271-\lambda^2$
1	270	$271-\lambda^2$
2	267	$271-\lambda^2$
3	262	$271-\lambda^2$
4	255	$271-\lambda^2$
5	246	$271-\lambda^2$
6	235	$271-\lambda^2$
7	222	$271-\lambda^2$
8	207	$271-\lambda^2$
9	190	$271-\lambda^2$
10	171	$C(29- \lambda , 2)$
11	153	$C(29- \lambda , 2)$
12	136	$C(29- \lambda , 2)$
13	120	$C(29- \lambda , 2)$
14	105	$C(29- \lambda , 2)$
15	91	$C(29- \lambda , 2)$

lambda	Population	Closed-form branch
16	78	C(29- lambda ,2)
17	66	C(29- lambda ,2)
18	55	C(29- lambda ,2)
19	45	C(29- lambda ,2)
20	36	C(29- lambda ,2)
21	28	C(29- lambda ,2)
22	21	C(29- lambda ,2)
23	15	C(29- lambda ,2)
24	10	C(29- lambda ,2)
25	6	C(29- lambda ,2)
26	3	C(29- lambda ,2)
27	1	C(29- lambda ,2)

Appendix E. Primitive Shell and q-Spectrum Summary

Quantity	Value
Primitive antipodal addresses	864
Observation rank	6
Compatibility nullity	858
q minimum	130.593523477839227
q maximum	1744.361292059535117
q mean	757.916780244175357
q standard deviation	325.034198848753931
Mean/trace(R)	45.000000000000000
Sum coordinate squares	[38880, 38880, 38880]

The complete list of 864 primitive addresses, q-values, and observation features is included in the machine-readable companion.

Appendix F. Reproducibility Equations

Cell3 node: $p_L=B \ L/2$ for L in $\{-1,0,1\}^3$.

Lucas Supergroup outer basis: $B_S=9B$.

Cellular f-vector: $f_k=C(3,k)9^k10^{(3-k)}$.

Total cellular polynomial: $F(t)=(10+9t)^3$.

Interior polynomial: $F_{int}(t)=(8+9t)^3$.

Boundary polynomial: $F_{boundary}(t)=F(t)-F_{int}(t)$.

Euler values: $F(-1)=1$, $F_{boundary}(-1)=2$.

Internal interfaces: $I(m)=3m^2(m-1)$.

Boundary states: $N_{boundary}(m)=24m^2+2$.

Unique closure: $N_{boundary}(m)-2=I(m)$ gives $m=9$.

Sheet population: coefficient of $(x^{-9}+\dots+x^9)^3$.

Shell nodes: $N_r=24r^2+2$; antipodal pairs $P_r=12r^2+1$.

Primitive antipodal shell directions: $12 \ J_2(r)$.

Primitive q observation: $q(s)=s^T R \ s$.

Compatibility dimension: $864 - \text{rank}(A) = 858$.

Reciprocal principal spans: $9/\sqrt{\rho_i}$.